TRUNKBELAY · USER REQUIREMENTS

TrunkBelay user requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First issue (kit 1.9.1 upgrade), drawn from the users, stakeholders and constraints of TKB-PRB-001 v0.4

TrunkBelay user requirements

What the people in a palm rescue need, written from their side. The product requirements that meet these needs are in TKB-PRS-001 (`03-requirements.md`) and the links are in the traceability matrix TKB-RTM-001 (`03b-traceability.md`). The needs are drawn from the reported rescues and the users and stakeholders in TKB-PRB-001; none has yet been confirmed with a climber, a fire crew or a trainer, so all are proposals for co-design. Method: STANDARDS section 22.

> **Safety:** These needs describe rescue equipment that holds a person at height. Meeting them on paper does not make the kit safe; each is confirmed by test at TRL 4.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Safety	Holds on a wet palm	The anchor holds the stranded person on a wet, fibrous, often curved palm trunk without slipping	High	Rescuer; stranded climber (PRB Table 2)
UR-002	Safety	Holds as the load moves	The anchor stays secure while the person's weight comes onto it and as they are lowered past it	High	Rescuer (PRB Table 2)
UR-003	Deployment and assembly	Fits the palms we climb	One kit fits the coconut palms climbed locally, about 200 to 450 mm across, with no tools	High	Rescuer; climbers' collective (PRB Tables 2 and 3)
UR-004	Usability	Fast at height	Once beside the victim, the rescuer can fix the anchor and take the victim's weight within a few minutes	High	Stranded climber; rescuer (PRB Table 2)
UR-005	Performance	Controlled lowering	One rescuer can lower the person smoothly, at walking pace or slower, without a strong grip	High	Rescuer (PRB Table 2)
UR-006	Performance	Reaches the ground	The rope reaches the ground from the tallest palms climbed, about 25 m	High	Rescuer; helpers on the ground (PRB Table 2)
UR-007	Logistics	Light to carry up	What the rescuer carries up the trunk is light and hangs from the harness, leaving both hands free to climb	High	Rescuer (PRB Table 2)
UR-008	Environment and durability	No harm to the palm	Using the kit does not cut or wound the palm	Medium	Palm owners and farmers (PRB Table 3)
UR-009	Cost	Affordable to a group	A climber group or a training scheme can afford a kit	Medium	Trainer; Coconut Development Board (PRB Tables 2 and 3)
UR-010	Usability	Learnable in half a day	Climbers new to the kit can run a supervised rescue drill after a half-day session	Medium	Trainer; fire crew (PRB Table 2)
UR-011	Safety	Holds if the rescuer lets go	If the rescuer is hit, faints or lets go, the person is still held	High	Stranded climber; rescuer (PRB Table 2)
UR-012	Safety	Head up once freed	Once freed from the climbing device, the person hangs head up and cannot slip out	High	Stranded climber (PRB Table 2)
UR-013	Environment and durability	Survives the climate	The kit survives heat, humidity, monsoon rain and storage in a shed	Medium	Climbers (PRB operating environment)
UR-014	Maintenance	Easy to inspect and renew	Parts that wear can be checked by eye and replaced locally	Low	Trainer; climbers' collective (PRB Table 3)
UR-015	Regulatory	Fits rescue practice	The kit uses rescue gear that fire crews recognise and is never presented as certified equipment	Medium	Fire and rescue crew (PRB Table 2)

ID	Category	Title	Need	Priority	Source
UR-016	Deployment and assembly	Made and mended locally	A local workshop can make and repair the made parts	Medium	Local fabricators; trainer (PRB Table 3)
UR-017	Regulatory	Free to build and share	Anyone can build, adapt and share the design without infringing others' patents	Medium	Design Molecule; co-design partners (PRB Table 3)

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.